



Year 12  
**Student Number**

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# 2020 HSC Trial Examination

## Year 12 Mathematics Extension 1

Examiner: AC

**Reading time – 10 minutes**

**Writing time – 2 hours**

### General Instructions

- Write using black or blue pen.
- Read the instructions carefully – you are required to answer the questions in the space provided.
- If you use booklets, start each question in a separate writing booklet.
- Write your student name clearly on each page.
- Board-approved calculators may be used, unless stated otherwise.
- All diagrams must be drawn in pencil.
- Do not remove this question paper from the examination room.

| Section           | Guidance                                                                                                                                                     | Marks Available | Your Score |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|
| <b>SECTION I</b>  | <ul style="list-style-type: none"><li>• Type of Questions – Multiple Choice</li><li>• Attempt Questions 1 - 10</li><li>• Timing 15 minutes</li></ul>         | 10              |            |
| <b>SECTION II</b> | <ul style="list-style-type: none"><li>• Type of Questions – Multiple Choice</li><li>• Attempt Questions 11 - 14</li><li>• Timing 1 hour 45 minutes</li></ul> | 60              |            |
| <b>Totals</b>     |                                                                                                                                                              | 70              |            |

|                   |      |   |
|-------------------|------|---|
| <b>FINAL MARK</b> | / 70 | % |
|-------------------|------|---|

Your examination paper begins overleaf.

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## Section 1: 10 marks

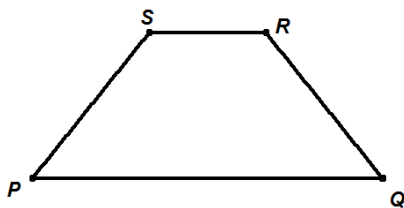
### Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

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- 1 Given  $(x - 2)$  is a factor of  $x^3 - 8x^2 + 21x - A$ , which of the following is the value of  $A$ ?
- (A)  $A = -82$
- (B)  $A = -2$
- (C)  $A = 2$
- (D)  $A = 18$
- 2 Which of the following is the derivative of  $\tan^{-1}(3x)$ ?
- (A)  $3 \tan^{-1} x$
- (B)  $\frac{3}{1+x^2}$
- (C)  $\frac{3}{1+9x^2}$
- (D)  $3 \sec^2 3x$
- 3  $PQRS$  is a trapezium where  $\overrightarrow{PS} = \vec{p}$ ,  $\overrightarrow{SR} = \vec{s}$  and  $\overrightarrow{PQ} = 2 \overrightarrow{SR}$ .

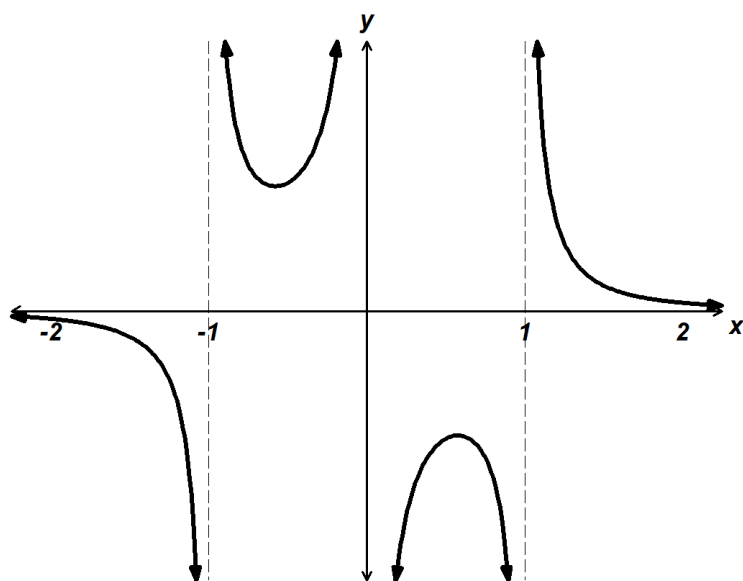


Which of the following is equivalent to  $\overrightarrow{QS}$ ?

- (A)  $2\vec{s} + \vec{p}$
- (B)  $2\vec{s} - \vec{p}$
- (C)  $\vec{p} - 2\vec{s}$
- (D)  $-\vec{p} - 2\vec{s}$

- 4 Which of the following is the coefficient of  $x^4$  in the expansion  $\left(x + \frac{3}{x}\right)^8$ ?
- (A) 28
- (B) 56
- (C) 84
- (D) 252

5

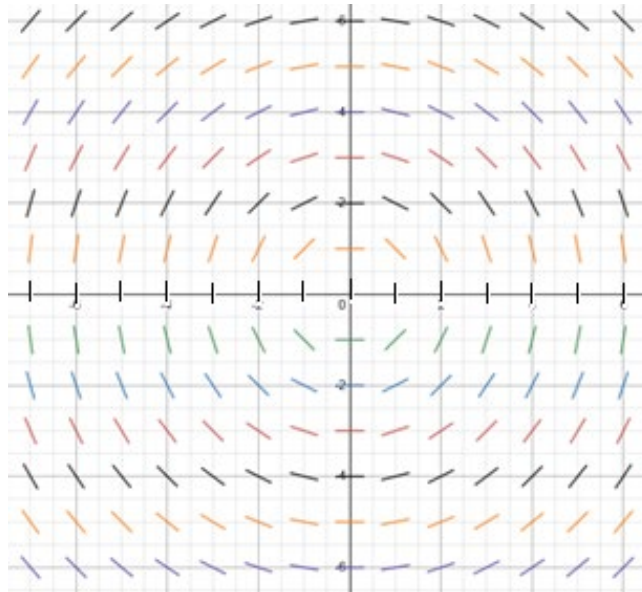


The graph above shows  $y = \frac{1}{f(x)}$ .

Which of the equations below best represents  $f(x)$ ?

- (A)  $f(x) = x^2 - 1$
- (B)  $f(x) = x(x^2 - 1)$
- (C)  $f(x) = x^2(x^2 - 1)$
- (D)  $f(x) = x^2(x^2 - 1)^2$

- 6 The slope field for a first order differential equation is shown below.



Which of the following could be the differential equation represented?

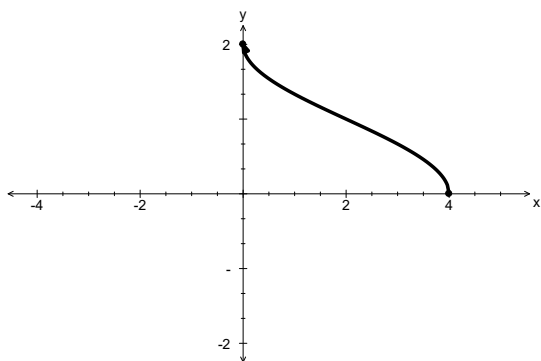
- (A)  $\frac{dy}{dx} = \frac{x}{y}$
- (B)  $\frac{dy}{dx} = \frac{-x}{y}$
- (C)  $\frac{dy}{dx} = xy$
- (D)  $\frac{dy}{dx} = -xy$

- 7 Four female and four male students are to be seated around a circular table.  
In how many ways can this be done if the males and females must alternate?

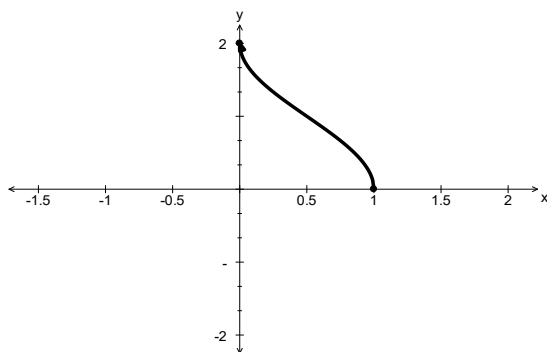
- (A)  $4! \times 4!$
- (B)  $3! \times 4!$
- (C)  $3! \times 3!$
- (D)  $2 \times 3! \times 3!$

8 Which of the graphs below shows  $y = 2 \cos^{-1}\left(\frac{x}{2} - 1\right)$ ?

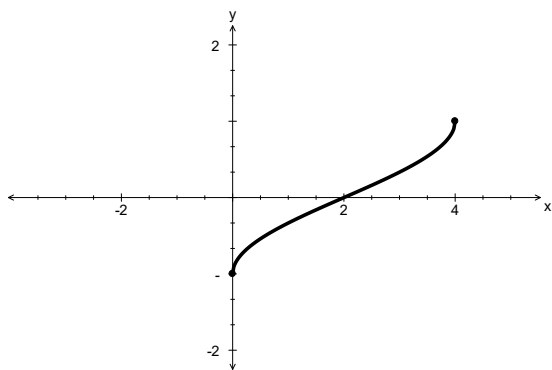
(A)



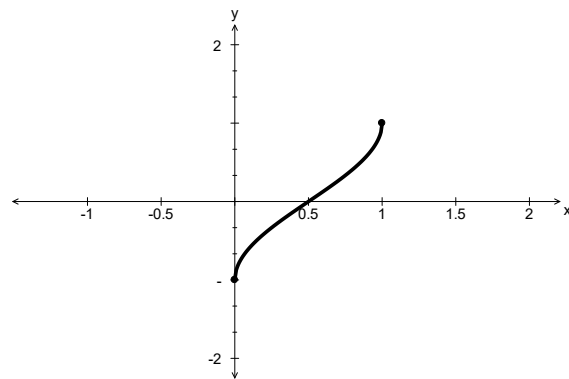
(B)



(C)



(D)



9 Which of the following expressions represents the area of the region bounded by the curve  $y = \sin^3 x$  and the  $x$ -axis from  $x = -\pi$  to  $x = 2\pi$ ? Use the substitution  $u = \cos x$ .

(A)  $-\int_{-\pi}^{2\pi} (1 - u^2) du$

(B)  $-3 \int_0^{\pi} (1 - u^2) du$

(C)  $-\int_{-1}^1 (1 - u^2) du$

(D)  $3 \int_{-1}^1 (1 - u^2) du$

- 10 Emma made an error proving that  $2^n + (-1)^{n+1}$  is divisible by 3 for all integers  $n \geq 1$  using mathematical induction. The proof is shown below.

Step 1: To prove  $2^n + (-1)^{n+1}$  is divisible by 3 ( $n$  is an integer)

To prove true for  $n = 1$

$$2^1 + (-1)^{1+1} = 2 + 1$$

$$= 3 \times 1 \quad \textbf{Line 1}$$

Result is true for  $n = 1$

Step 2: Assume true for  $n = k$

$$\text{ie. } 2^k + (-1)^{k+1} = 3m \text{ (} m \text{ is an integer)} \quad \textbf{Line 2}$$

Step 3: To prove true for  $n = k + 1$

$$2^{k+1} + (-1)^{k+1+1} = 2(2^k) + (-1)^{k+2} \quad \textbf{Line 3}$$

$$= 2[3m + (-1)^{k+1}] + (-1)^{k+2} \quad \textbf{Line 4}$$

$$= 2 \times 3m + 2 \times (-1)^{k+2} + (-1)^{k+2}$$

$$= 3[2m + (-1)^{k+2}]$$

Which is a multiple of 3 since  $m$  and  $k$  are integers.

Step 4: True by induction

In which line did Emma make an error?

- (A) Line 1
- (B) Line 2
- (C) Line 3
- (D) Line 4

## **Section II: 60 marks**

### **Attempt Questions 11 – 14**

**Allow about 1 hour and 45 minutes for this section**

Answer each question in a separate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

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#### **Question 11 (15 marks)    *Start a new writing booklet***

- (a) Consider the function  $f(x) = x^2 - 4x + 6$ .
- (i) Explain why the domain of  $f(x)$  must be restricted if  $f(x)$  is to have an inverse function. 1
- (ii) Given that the domain of  $f(x)$  is restricted to  $x \leq 2$ , find an expression for  $f^{-1}(x)$ . 2
- (iii) Given the restriction in part (ii), sketch  $y = f^{-1}(x)$ . 2
- (iv) The curve  $y = f(x)$  with its restricted domain and the curve  $y = f^{-1}(x)$  intersect at point  $P$ .  
Find the coordinates of  $P$ . 1
- (b) Use the substitution  $u = 1 + 2 \tan x$  to evaluate  $\int_0^{\frac{\pi}{4}} \frac{1}{(1 + 2 \tan x)^2 \cos^2 x} dx$ . 3

*(Q11 continues on the next page)*

*(Q11 continued)*

- (c) Solve the equation  $\cos x - \sin x = 1$ , where  $0 \leq x \leq 2\pi$ . **3**

- (d) The column (position) vector notation of 4 vectors is shown below.

$$P = \begin{pmatrix} -8 \\ -8 \end{pmatrix} \quad Q = \begin{pmatrix} 3 \\ 6 \end{pmatrix} \quad R = \begin{pmatrix} 1 \\ 5 \end{pmatrix} \quad S = \begin{pmatrix} -5 \\ 7 \end{pmatrix}$$

Find the column (position) vector notation of:

(i)  $\overrightarrow{PQ}$  **1**

(ii)  $\overrightarrow{RS}$  **1**

(iii)  $-\overrightarrow{PQ} - \overrightarrow{RS}$  **1**

**End of Question 11. Start a New Booklet.**

**Question 12** (15 marks) *Start a new writing booklet*

- (a) A particle is moving in a straight line such that its displacement ( $x$  metres) from a fixed point  $O$  after  $t$  seconds is given by  $x = \cos 2t + \sqrt{3} \sin 2t$ .
- (i) What is the maximum distance of the particle from  $O$ ? 2
- (ii) When is the particle first at the origin? 1
- (b) A heated metal ball is dropped into a liquid. As the ball cools, its temperature,  $T$  °C,  $t$  minutes after it enters the liquid, is given by:

$$T = 400e^{-0.05t} + 25, \quad t \geq 0$$

- (i) Find the temperature of the ball as it enters the liquid. 1
- (ii) Find the value of  $t$  if  $T = 300$ . Answer correct to 3 significant figures. 1
- (iii) Find the rate at which the temperature of the ball is decreasing at the instant when  $t = 50$ . Give your answer in °C per minute to 3 significant figures. 2
- (iv) Using the equation for temperature  $T$  in terms of  $t$ , given above, to explain why the temperature of the ball can never fall to 20°C. 1

(c) Find  $\int_0^{\pi} \frac{4}{\sqrt{16-x^2}} dx$ . 2

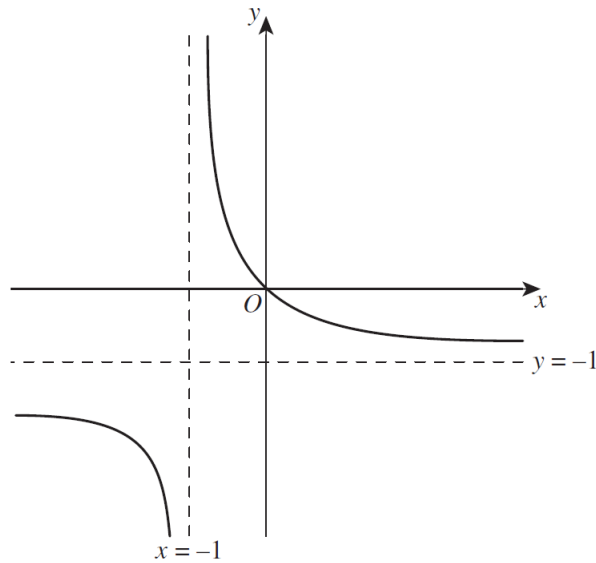
(d) (i) Use the substitution  $t = \tan \frac{x}{2}$  to show that  $\operatorname{cosec} x + \cot x = \cot \frac{x}{2}$ . 2

(ii) Hence evaluate  $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\operatorname{cosec} x + \cot x) dx$ . Answer in simplest exact form. 3

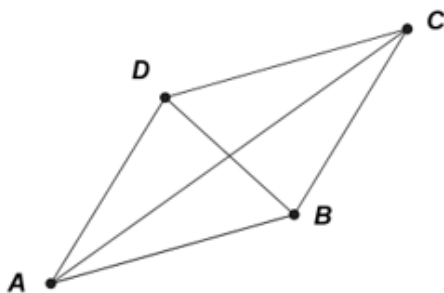
**End of Question 12.** *Start a New Booklet.*

**Question 13** (15 marks) *Start a new writing booklet*

- (a) The diagram below is the sketch of the graph of the function  $f(x) = -\frac{x}{x+1}$ .



- (i) Sketch the graph of  $y = (f(x))^2$ , showing all asymptotes and intercepts. 2
- (ii) Solve the equation  $(f(x))^2 = f(x)$ . 1
- (b)  $ABCD$  is a rhombus with  $\overrightarrow{AB} = \mathbf{a}$  and  $\overrightarrow{AD} = \mathbf{d}$ .



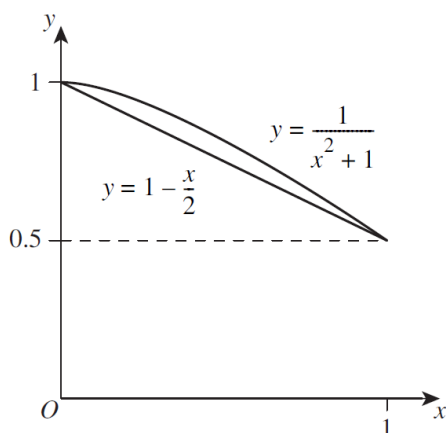
Use vector methods to prove that the diagonals of the rhombus are perpendicular to each other.

2

*(Q13 continues on the next page)*

(Q13 continued)

- (c) The diagram shows the graph of  $y = \frac{1}{x^2 + 1}$  and the graph of  $y = 1 - \frac{x}{2}$  for  $0 \leq x \leq 1$ .



- (i) Find the exact volume of the solid of revolution formed when the region bounded by the graph of  $y = \frac{1}{x^2 + 1}$ , the  $y$ -axis and the line  $y = \frac{1}{2}$  is rotated about the  $y$ -axis. 2
- (ii) Find the exact volume of the solid of revolution formed when the region bounded by the graph of  $y = 1 - \frac{x}{2}$ , the  $y$ -axis and the line  $y = \frac{1}{2}$  is rotated about the  $y$ -axis. 2
- (iii) Use the results from parts (c)(i) and (c)(ii) to show that  $\frac{2}{3} < \ln 2$ . 1
- (d) A multiple-choice test contains ten questions. Each question has four choices for the correct answer. Only one of the choices is correct.
- (i) What is the probability of getting 70% correct with random guessing? 1
- (ii) What is the probability of getting at most 70% correct with random guessing? 2
- (e) A binomial random variable  $X$  has a mean of 15 and a variance of 10.  
What are the parameters  $n$  and  $p$ ? 2

**End of Question 13.** *Start a New Booklet.*

**Question 14** (15 marks) *Start a new writing booklet*

- (a) Prove by mathematical induction that, for all integers  $n \geq 1$ ,

$$\frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{n(n+2)} = \frac{3}{2} - \frac{2n+3}{(n+1)(n+2)}.$$

3

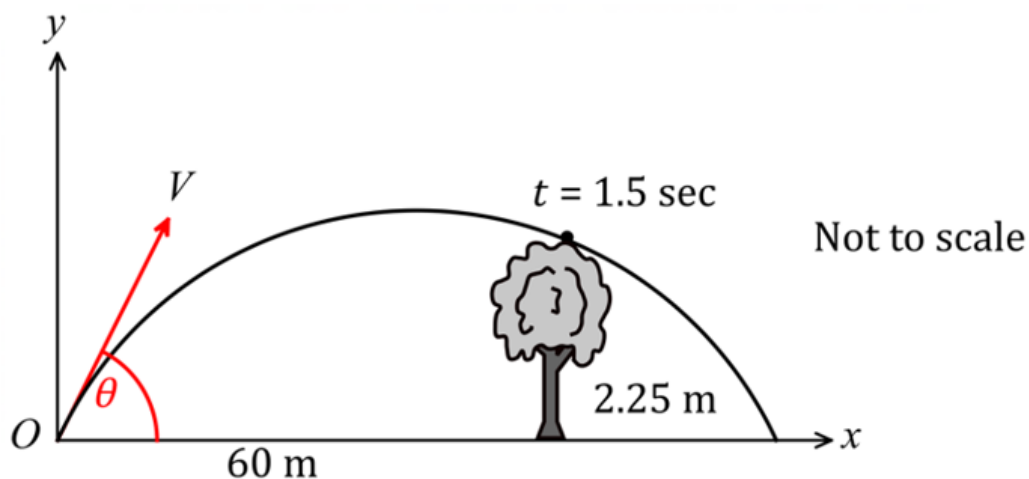
- (b) A bag contains  $n$  red marbles and one blue marble. Three marbles are drawn

(without replacement). The probability that the three marbles are red is  $\frac{5}{8}$ .

Find the value of  $n$ .

2

- (c) A golfer hits a golf ball from a point  $O$  with speed  $V \text{ ms}^{-1}$  at an angle  $\theta^\circ$  above the horizontal, where  $0 < \theta < \frac{\pi}{2}$ . The ball just passes over a 2.25 m high tree after 1.5 seconds. The tree is 60 metres away from the point from which the ball was hit. Assume  $g = 10 \text{ ms}^{-2}$ .



- (i) What is the angle of projection of the golf ball to the nearest minute?  
Assume the horizontal and vertical displacements of the golf ball are given by the vector  $\underline{r}(t) = (Vt \cos \theta)\mathbf{i} + (-5t^2 + Vt \sin \theta)\mathbf{j}$ .

3

- (ii) What is the initial speed ( $V \text{ ms}^{-1}$ ) of the golf ball, correct to the nearest whole number?

1

(Q14 continues on the next page)

(Q14 continued)

- (d) The population,  $P$ , of animals in an environment in which there are scarce resources is increasing such that  $\frac{dP}{dt} = P(100 - P)$ , where  $t$  is time.

When  $t = 0$ ,  $P = 10$ .

(i) Show that  $\frac{1}{100} \left( \frac{1}{P} + \frac{1}{100 - P} \right) = \frac{1}{P(100 - P)}$ . 1

(ii) Find an expression for  $P$  in terms of  $t$ . 3

- (e) The table shows selected values of a one-to-one differentiable function  $g(x)$  and its derivative  $g'(x)$ .

|         |      |               |
|---------|------|---------------|
| $x$     | $-1$ | $0$           |
| $g(x)$  | $-5$ | $-1$          |
| $g'(x)$ | $3$  | $\frac{1}{2}$ |

Let  $f(x)$  be a function such that  $f(x) = g^{-1}(x)$ .

Find the value of  $f'(-1)$ . 2

**End of Question 14.**

**End of Examination.**

**NGS Trial Examination 2020**  
**Year 12 Mathematics Extension 1**  
**Worked solutions and marking guidelines**

**Section 1**

**10 marks**

**Question 1 (1 mark)**

| Solution                                                                                         | Answer   | Mark     |
|--------------------------------------------------------------------------------------------------|----------|----------|
| $(x-2)$ is a factor of<br>$x^3 - 8x^2 + 21x - A$<br>$\therefore 8 - 32 + 42 - A = 0$<br>$A = 18$ | <b>D</b> | <b>1</b> |

**Question 2 (1 mark)**

| Solution                                                                                        | Answer   | Mark     |
|-------------------------------------------------------------------------------------------------|----------|----------|
| If $y = \tan^{-1}(3x)$<br>$\frac{dy}{dx} = \frac{1}{1+(3x)^2} \times 3$<br>$= \frac{3}{1+9x^2}$ | <b>C</b> | <b>1</b> |

**Question 3 (1 mark)**

| Solution                                                                                    | Answer   | Mark     |
|---------------------------------------------------------------------------------------------|----------|----------|
| $\overrightarrow{QS} = \overrightarrow{PS} - \overrightarrow{PQ}$<br>$= \vec{p} - 2\vec{s}$ | <b>C</b> | <b>1</b> |

**Question 4 (1 mark)**

| Solution                                                                                                                                                                                                          | Answer   | Mark     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|
| $T_{k+1} = {}^8C_k (x)^{8-k} \left(\frac{3}{x}\right)^k$<br>$= {}^8C_k x^{8-k} (3^k x^{-k})$<br>$= {}^8C_k (3)^k x^{8-2k}$<br>$x^4 \Rightarrow 8 - 2k = 4$<br>$k = 2$<br>the term is ${}^8C_2 \times (3)^2 = 252$ | <b>D</b> | <b>1</b> |

**Question 5 (1 mark)**

| Solution                                                | Answer   | Mark     |
|---------------------------------------------------------|----------|----------|
| By inspection and properties<br>of $y = \frac{1}{f(x)}$ | <b>B</b> | <b>1</b> |

**Question 6 (1 mark)**

| Solution                                                                                                          | Answer   | Mark     |
|-------------------------------------------------------------------------------------------------------------------|----------|----------|
| By considering slopes at<br>different points of cartesian<br>plane and testing with each<br>differential equation | <b>B</b> | <b>1</b> |

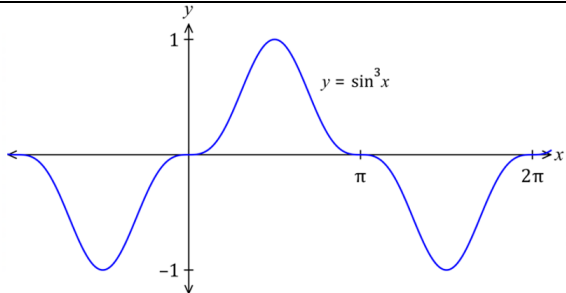
**Question 7 (1 mark)**

| Solution                                                                                                       | Answer   | Mark     |
|----------------------------------------------------------------------------------------------------------------|----------|----------|
| Seat females in $(4-1)!$ ways.<br>Then seat males in $4!$ ways<br>$\therefore$ number of ways $= 3! \times 4!$ | <b>B</b> | <b>1</b> |

**Question 8 (1 mark)**

| Solution                                                                                                                                                 | Answer   | Mark     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|
| For $y = 2 \cos^{-1} \left( \frac{x}{2} - 1 \right)$<br>Range $0 \leq y \leq 2\pi$<br>Domain is $D: -1 \leq \frac{x}{2} - 1 \leq 1$<br>$0 \leq x \leq 4$ | <b>A</b> | <b>1</b> |

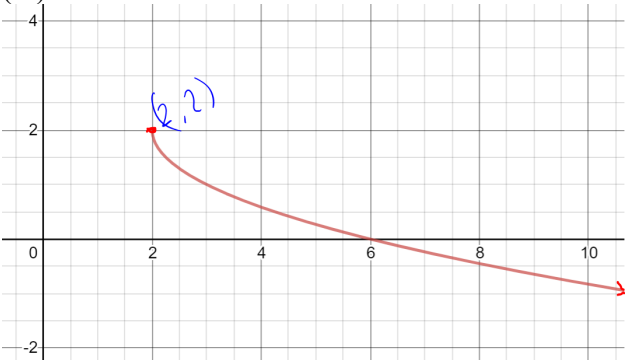
**Question 9 (1 mark)**

| Solution                                                                                                                                                                                                                                                                                                                                                                                                                                | Answer   | Mark     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|
|  <p>There are 3 equivalent areas from <math>x = -\pi</math> to <math>x = 2\pi</math></p> $u = \cos x$ $\frac{du}{dx} = -\sin x$ $du = -\sin x dx$ $x = 0, u = 1 \text{ and } x = \pi, u = -1$ $A = 3 \times \int_0^\pi \sin^3 x dx$ $= -3 \int_0^\pi (1 - \cos^2 x) \times -\sin x dx$ $= -3 \int_1^{-1} (1 - u^2) du$ $= 3 \int_{-1}^1 (1 - u^2) du$ | <b>D</b> | <b>1</b> |

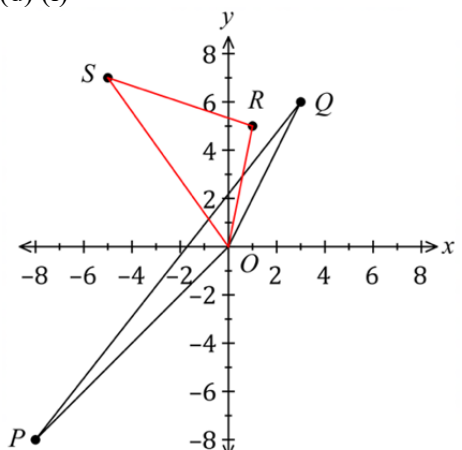
**Question 10 (1 mark)**

| Solution                                                                                                                                                                                                                                                                                                                    | Answer   | Mark     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|
| Step 3: To prove true for $n = k + 1$<br>$2^{k+1} + (-1)^{k+1+1} = 2(2^k) + (-1)^{k+2}$<br>$= 2[3m - (-1)^{k+1}] + (-1)^{k+2}$ <b>Error Line 4</b><br>$= 2 \times 3m - 2 \times (-1)^{k+1} - (-1)^{k+1}$<br>$= 3[2m - (-1)^{k+1}]$<br>Which is a multiple of 3 since $m$ and $k$ are integers.<br>Step 4: True by induction | <b>D</b> | <b>1</b> |

**Q11 (15 mark)**

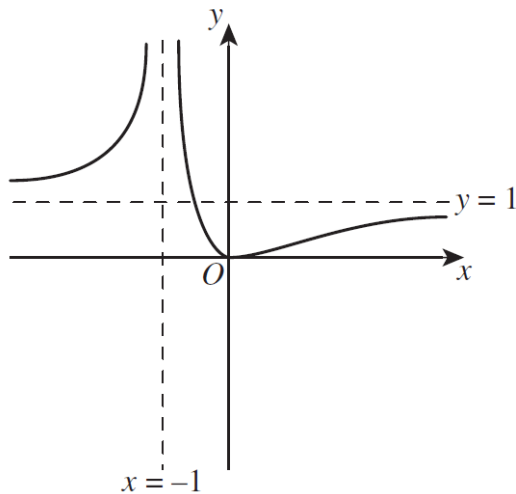
| <b>Solution</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Mark (Guide only)</b>                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <p>(a) (i) <math>f(x) = x^2 - 4x + 6</math> is a parabola. Excluding the turning point at <math>(2, 2)</math>, for each value of <math>f(x)</math> in the range there are two <math>x</math>-values. Geometrically, this corresponds to a horizontal line intersecting the graph twice.</p> <p>If <math>x</math> and <math>y</math> are swapped, each <math>x</math>-value in the domain will have two <math>y</math>-values. Hence the inverse will not be a function.</p>                                                                                                                           | 1 Mark: Explains using the horizontal line test or equivalent merit.                                           |
| <p>(ii) Use the completing the square method to express <math>f(x)</math> in turning point form:</p> $f(x) = x^2 - 4x + 6$ $= (x - 2)^2 + 2 \quad (x \geq 2)$ <p>Swap <math>x</math> and <math>y</math>, then make <math>y</math> the subject.</p> $x = (y - 2)^2 + 2$ $x - 2 = (y - 2)^2$ $y - 2 = -\sqrt{x - 2} \quad (\sqrt{x - 2} \text{ is discarded as } y \leq 2)$ $y = -\sqrt{x - 2} + 2$ $f^{-1}(x) = -\sqrt{x - 2} + 2 \quad (x \geq 2)$                                                                                                                                                    | <p>2 Marks: Correct Answer</p> <p>1 Mark: Swaps <math>x</math> and <math>y</math> OR equivalent merit.</p>     |
| <p>(iii)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <p>2 Marks: Correct shape and start at <math>(2, 2)</math></p> <p>1 Mark: Correct shape OR starting point.</p> |
| <p>(iv) The curves <math>y = f(x)</math> and <math>y = f^{-1}(x)</math> have a common intersection with the line <math>y = x</math>.</p> <p>For example, attempting to solve <math>f(x) = x</math> for <math>x</math>:</p> $x^2 - 4x + 6 = x$ $x^2 - 5x + 6 = 0$ $x = 2, 3$ <p>When <math>x = 2</math>, <math>y = 2</math> and so <math>(2, 2)</math> lies on the line <math>y = x</math>.</p> <p>When <math>x = 3</math>, <math>y = 1</math> and so <math>(3, 1)</math> does not lie on the line <math>y = x</math>.</p> <p>Therefore the coordinates of <math>P</math> are <math>(2, 2)</math>.</p> | 1 Mark: Correct solution                                                                                       |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(b) Let <math>u = 1 + 2 \tan x</math>.</p> $\frac{du}{dx} = 2 \sec^2 x = \frac{2}{\cos^2 x} \Rightarrow dx = \frac{\cos^2 x}{2} du$ <p>When <math>x = 0</math>, <math>u = 1</math> and when <math>x = \frac{\pi}{4}</math>, <math>u = 3</math>.</p> $\int_0^{\frac{\pi}{4}} \frac{1}{(1 + 2 \tan x)^2 \cos^2 x} dx = \int_1^3 \frac{1}{2u^2} du$ $= -\left[\frac{1}{2u}\right]_1^3$ $= -\left(\frac{1}{6} - \frac{1}{2}\right)$ $= \frac{1}{3}$                                                                                                                                                                                                                                                                                                                                                                                            | <p>3 Marks: Correct solution<br/>                 2 Marks: Finds expression for integral in terms of u, or equivalent merit.<br/>                 1 Mark: Derives <math>u=1+2\tan x</math> correctly</p>                                                   |
| <p>(c) Substituting <math>\cos x = \frac{1-t^2}{1+t^2}</math>, <math>\sin x = \frac{2t}{1+t^2}</math> where <math>t = \tan \frac{1}{2}x</math> into <math>\cos x - \sin x = 1</math> and expressing</p> $1 = \frac{1+t^2}{1+t^2} \text{ gives:}$ $\frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2} = \frac{1+t^2}{1+t^2}$ $\frac{1-t^2-2t-1-t^2}{1+t^2} = 0$ $\frac{-2(t^2+t)}{1+t^2} = 0$ $t^2+t=0$ $t(t+1)=0$ $t=-1, 0$ <p><math>\tan \frac{1}{2}x = -1, 0</math></p> <p><math>\tan \frac{1}{2}x = 0 \Rightarrow \frac{1}{2}x = 0, \pi</math></p> <p><math>\tan \frac{1}{2}x = -1</math></p> <p><math>\tan</math> is negative in the second quadrant and the related angle is <math>\frac{\pi}{4}</math>.</p> <p><math>\tan \frac{1}{2}x = -1 \Rightarrow \frac{x}{2} = \frac{3\pi}{4}</math></p> <p>So <math>x = 0, \frac{3\pi}{2}, 2\pi</math>.</p> | <p>3 Marks: Correct solution<br/>                 2 Marks: Determines that <math>\tan(\frac{1}{2}x) = -1, 0</math><br/>                 1 Mark: Attempts to form a quadratic equation in <math>t</math> with some correct working OR equivalent merit.</p> |

|                                                                                                                                                                                   |                                |
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| <p>(d) (i)</p>  <p><math>\overrightarrow{PQ} = \begin{pmatrix} 11 \\ 14 \end{pmatrix}</math></p> | <p>1 Mark: Correct answer.</p> |
| <p>(ii)</p> <p><math>\overrightarrow{RS} = \begin{pmatrix} -6 \\ 2 \end{pmatrix}</math></p>                                                                                       | <p>1 Mark: Correct answer.</p> |
| <p>(iii)</p> <p><math>-\overrightarrow{PQ} - \overrightarrow{RS} = \begin{pmatrix} -5 \\ -16 \end{pmatrix}</math></p>                                                             | <p>1 Mark: Correct answer.</p> |

| Q12 (15 mark)                                                                                                                                                                                                                                  |                                                                                                                                                           |
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| <p>(a) (i) Using auxiliary angle:</p> $x = \cos 2t + \sqrt{3} \sin 2t = 2 \cos \left( 2t - \frac{\pi}{3} \right)$ <p><math>\therefore</math> maximum distance from <math>O = 2</math> metres</p>                                               | <p>2 marks: Correct Answer<br/>1 Mark: Some attempt at Auxiliary Angle with some correct working OR equivalent merit</p>                                  |
| <p>(ii)</p> $2 \cos \left( 2t - \frac{\pi}{3} \right) = 0$ $2t - \frac{\pi}{3} = \frac{\pi}{2}$ $t = \frac{5\pi}{12} \text{ seconds}$                                                                                                          | <p>1 Mark: Correct answer.</p>                                                                                                                            |
| <p>(b) (i)</p> <p>Ball enters the liquid when <math>t = 0</math></p> $T = 400e^{-0.05t} + 25$ $= 400e^{-0.05 \times 0} + 25 = 425^\circ\text{C}$                                                                                               | <p>1 Mark: Correct answer.</p>                                                                                                                            |
| <p>(ii)</p> $300 = 400e^{-0.05t} + 25$ $e^{-0.05t} = \frac{275}{400}$ $-0.05t = \ln \left( \frac{11}{16} \right)$ $t = 7.4938... \approx 7.49 \text{ min (3 sig. fig.)}$                                                                       | <p>1 Mark: Correct answer.</p>                                                                                                                            |
| <p>(iii)</p> $T = 400e^{-0.05t} + 25$ $\frac{dT}{dt} = -20e^{-0.05t}$ $= -20e^{-0.05 \times 50}$ $= -1.6416... \approx -1.64^\circ\text{C/min}$ <p><math>\therefore</math> Rate of decrease is <math>1.64^\circ\text{C}</math> per minute.</p> | <p>2 Marks: Correct answer.<br/>1.5 marks: Only giving <math>-1.64/\text{min}</math><br/>1 Mark: Differentiates correctly to find the rate of change.</p> |
| <p>(iv)</p> <p>When <math>t</math> approaches infinity then <math>e^{-0.05t} \rightarrow 0</math><br/> <math>\therefore T &gt; 25</math> and can never fall to <math>20^\circ\text{C}</math>.</p>                                              | <p>1 Mark: Correct answer.</p> <p><b>(Must <u>use the equation</u> to show correctly)</b></p>                                                             |
| (c)                                                                                                                                                                                                                                            |                                                                                                                                                           |

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| $\int_0^{\pi} \frac{4}{\sqrt{16-x^2}} dx = 4 \left[ \sin^{-1} \left( \frac{x}{4} \right) \right]_0^{\pi}$ $= 4 \left[ \sin^{-1} \left( \frac{\pi}{4} \right) - \sin^{-1}(0) \right]$ $= 4 \left[ \frac{1}{\sqrt{2}} - 0 \right]$ $= \frac{4}{\sqrt{2}} = 2\sqrt{2}$                                                                                                                                                                                                                                                                                                                                | <p>2 Marks: Correct answer.</p> <p>1 Mark: Finds the correct integration.</p>                                                                              |
| <p>(d) (i)</p> $\text{LHS} = \operatorname{cosec} x + \cot x$ $= \frac{1+t^2}{2t} + \frac{1-t^2}{2t}$ $= \frac{1+t^2+1-t^2}{2t}$ $= \frac{1}{t}$ $= \cot \frac{x}{2}$ $= \text{RHS}$                                                                                                                                                                                                                                                                                                                                                                                                               | <p>2 Marks: Correct answer.</p> <p>1 Mark: Writes cosec x and cot x in terms of t.</p>                                                                     |
| <p>(ii)</p> $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} (\operatorname{cosec} x + \cot x) dx = \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cot \frac{x}{2} dx$ $= 2 \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \frac{0.5 \cos \frac{x}{2}}{\sin \frac{x}{2}} dx$ $= 2 \left[ \ln \left( \sin \frac{x}{2} \right) \right]_{\frac{\pi}{3}}^{\frac{\pi}{2}}$ $= 2 \left[ \ln \left( \sin \frac{\pi}{4} \right) - \ln \left( \sin \frac{\pi}{6} \right) \right]$ $= 2 \left( \ln \frac{1}{\sqrt{2}} - \ln \frac{1}{2} \right)$ $= 2 \ln \left( 2^{-\frac{1}{2}} \div 2^{-1} \right)$ $= 2 \ln 2^{\frac{1}{2}}$ $= \ln 2$ | <p>3 Marks: Correct answer.</p> <p>2 Marks: Makes significant progress.<br/><u>OR</u> uses logs correctly</p> <p>1 Mark: Finds the primitive function.</p> |

**Q13 (15 mark)****(a) (i)**

2 Marks: Sketches correct graph with asymptotes at  $x=-1$  and  $y=1$   
 1 Mark: Shows min turning point at origin  
 OR equivalent merit.

**(ii)**

$$(f(x))^2 = f(x) \Rightarrow f(x)(f(x) - 1) = 0$$

So  $f(x) = 1$  or  $f(x) = 0$ .

$$-\frac{x}{x+1} = 1 \Rightarrow x = -\frac{1}{2}$$

Hence  $x = -\frac{1}{2}$  or  $x = 0$ .

**OR**

The graphs of  $y = f(x)$  and  $y = (f(x))^2$  intersect at  $O$ , where  $x = 0$ .

The graphs of  $y = f(x)$  and  $y = (f(x))^2$  intersect on the line  $y = 1$ , where  $x = -\frac{1}{2}$ .

1 Mark: Correct solution

**(b)**

$$\overrightarrow{AC} = \mathbf{a} + \mathbf{d}, \overrightarrow{DB} = \mathbf{a} - \mathbf{d}$$

$$\overrightarrow{AC} \cdot \overrightarrow{DB} = (\mathbf{a} + \mathbf{d}) \cdot (\mathbf{a} - \mathbf{d})$$

$$= \mathbf{a} \cdot \mathbf{a} - \mathbf{d} \cdot \mathbf{d}$$

$$= |\mathbf{a}|^2 - |\mathbf{d}|^2$$

$$= 0 \text{ (sides of rhombus have equal length)}$$

$$\text{But } \overrightarrow{AC} \cdot \overrightarrow{DB} = |\mathbf{a}| |\mathbf{d}| \cos \theta \text{ (where } \theta \text{ is the angle between the vectors)}$$

$$\therefore \cos \theta = 0$$

$$\theta = 90^\circ$$

*i.e.* diagonals are perpendicular

2 marks: Correct Answer.

1 mark: Writes  $\overrightarrow{AC} = \mathbf{a} + \mathbf{d}$

**(c)**

2 marks: Correct Answer.

1 mark: Gives correct integral for volume of revolution

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                             |
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| <p>(i) Rearranging <math>y = \frac{1}{x^2 + 1}</math> to express <math>x^2</math> in terms of <math>y</math> gives <math>x^2 = \frac{1}{y} - 1</math>.</p> $V = \pi \int_{\frac{1}{2}}^1 \left( \frac{1}{y} - 1 \right) dy$ $= \pi \left[ \ln y  - y \right]_{\frac{1}{2}}^1$ $= \pi \left( \ln 1 - 1 - \left( \ln \frac{1}{2} - \frac{1}{2} \right) \right)$ $= \pi \left( \ln 2 - \frac{1}{2} \right)$                                                                                                                                                                                                 |                                                                                             |
| <p>(ii) Rearranging <math>y = 1 - \frac{x}{2}</math> to express <math>x</math> in terms of <math>y</math> gives <math>x = 2(1 - y)</math>.</p> $V = \pi \int_{\frac{1}{2}}^1 (4(1 - y)^2) dy$ $= -\frac{4\pi}{3} \left[ (1 - y)^3 \right]_{\frac{1}{2}}^1$ $= -\frac{4\pi}{3} \left( 0 - \frac{1}{8} \right)$ $= \frac{\pi}{6}$ <p>Alternatively:</p> <p>The solid formed is a cone of radius 1 and height <math>\frac{1}{2}</math>.</p> <p>Substituting these values into <math>V = \frac{1}{3}\pi r^2 h</math> gives:</p> $V = \frac{1}{3} \times \pi \times 1^2 \times \frac{1}{2}$ $= \frac{\pi}{6}$ | <p>2 marks: Correct Answer.<br/>1 mark: Gives correct integral for volume of revolution</p> |
| <p>(iii) From the diagram, it can be reasoned that <math>\pi \left( \ln 2 - \frac{1}{2} \right) &gt; \frac{\pi}{6}</math>.</p> <p>So <math>\ln 2 - \frac{1}{2} &gt; \frac{1}{6} \Rightarrow \ln 2 &gt; \frac{2}{3}</math>.</p>                                                                                                                                                                                                                                                                                                                                                                           | <p>1 Mark: Correct answer.</p>                                                              |
| <p>(d) (i) Let <math>p</math> be the probability of getting the correct answer.<br/><math>p = \frac{1}{4}, n = 10</math></p> $P(X = x) = {}^{10}C_x \left( \frac{1}{4} \right)^x \left( \frac{3}{4} \right)^{10-x}$ $P(X = 7) = {}^{10}C_7 \left( \frac{1}{4} \right)^7 \left( \frac{3}{4} \right)^{10-7}$ $= \frac{405}{131072}$                                                                                                                                                                                                                                                                        | <p>1 Mark: Correct answer.</p>                                                              |

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| <p>(ii)</p> $P(X \leq 7) = 1 - (P(8) + P(9) + P(10))$ $= 1 - \left( {}^{10}C_8 \left(\frac{1}{4}\right)^8 \left(\frac{3}{4}\right)^{10-8} + {}^{10}C_9 \left(\frac{1}{4}\right)^9 \left(\frac{3}{4}\right)^{10-9} + {}^{10}C_{10} \left(\frac{1}{4}\right)^{10} \left(\frac{3}{4}\right)^{10-10} \right)$ $= 0.99958...$                                                                                                      | <p>2 Marks: Correct answer.<br/>1 Mark: Uses the complementary event or shows some understanding.</p>   |
| <p>(e)</p> $E(X) = np = 15 \text{ (1)}$ $Var(X) = np(1 - p) = 10 \text{ (2)}$ <p>Substituting equation (1) into (2)</p> $15 \times (1 - p) = 10$ $1 - p = \frac{10}{15} = \frac{2}{3} \text{ or } p = \frac{1}{3}$ <p>Substituting <math>p = \frac{1}{3}</math> into equation (1)</p> $n \times \frac{1}{3} = 15$ $n = 45$ <p><math>\therefore</math> Parameters are <math>n = 45</math> and <math>p = \frac{1}{3}</math></p> | <p>2 Marks: Correct answer.</p> <p>1 Mark: Finds one of the parameters or shows some understanding.</p> |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                           |
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| <p><b>Q14 (15 mark)</b></p> <p>(a)</p> <p>Consider <math>n = 1</math>.</p> $\text{LHS} = \frac{2}{1 \times 3} = \frac{2}{3} \text{ and}$ $\text{RHS} = \frac{3}{2} - \frac{2(1)+3}{(1+1)(1+2)} = \frac{4}{6} = \frac{2}{3} = \text{LHS.}$ <p>The statement is true when <math>n = 1</math>.</p> <p>Suppose true for <math>n = k</math>.</p> <p>So <math>\frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{k(k+2)} = \frac{3}{2} - \frac{2k+3}{(k+1)(k+2)}</math>.</p> <p>Show it is true for <math>n = k + 1</math>; that is,</p> $\frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{k(k+2)} + \frac{2}{(k+1)(k+3)} =$ $\frac{3}{2} - \frac{2(k+1)+3}{((k+1)+1)((k+1)+2)}$ $\text{LHS} = \frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{k(k+2)} + \frac{2}{(k+1)(k+3)}$ $= \frac{3}{2} - \frac{2k+3}{(k+1)(k+2)} + \frac{2}{(k+1)(k+3)}$ $= \frac{3}{2} - \frac{(2k+3)(k+3) - 2(k+2)}{(k+1)(k+2)(k+3)}$ $= \frac{3}{2} - \frac{2k^2 + 7k + 5}{(k+1)(k+2)(k+3)}$ $= \frac{3}{2} - \frac{(2k+5)(k+1)}{(k+1)(k+2)(k+3)}$ $= \frac{3}{2} - \frac{2k+5}{(k+2)(k+3)}$ $= \frac{3}{2} - \frac{2(k+1)+3}{((k+1)+1)((k+1)+2)}$ $= \text{RHS}$ <p>If true for <math>n = k</math>, then true for <math>n = k + 1</math>.</p> <p>Hence, by mathematical induction, true for <math>n \geq 1</math>.</p> | <p>3 Marks: Correct proof<br/>2 Marks: Establishes the inductive step<br/>OR equivalent merit.<br/>1 Mark: Establishes the <math>n=1</math> case or equivalent merit.</p> |
| <p>(b)</p> <p>Prob(3 red marbles drawn) = <math>\frac{{}^nC_3}{{}^{n+1}C_3}</math></p> <p style="text-align: center;">↓</p> <div style="display: flex; justify-content: space-between;"> <div style="width: 45%;"> <math display="block">\frac{n}{n+1} \times \frac{n-1}{n} \times \frac{n-2}{n-1} = \frac{5}{8}</math> <math display="block">\frac{n-2}{n+1} = \frac{5}{8}</math> <math display="block">n = 7</math> </div> <div style="width: 45%;"> <p style="text-align: center;">↓</p> <math display="block">\frac{{}^nC_3}{{}^{n+1}C_3} = \frac{5}{8}</math> <math display="block">\frac{\left( \frac{n(n-1)(n-2)}{3 \times 2 \times 1} \right)}{\frac{(n+1)(n)(n-1)}{3 \times 2 \times 1}} = \frac{5}{8}</math> <math display="block">\frac{n-2}{n+1} = \frac{5}{8}</math> <math display="block">8n - 16 = 5n + 5</math> <math display="block">n = 7</math> </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <p>2 marks: Correct Answer<br/>1 mark: Writes correct answer for probability</p>                                                                                          |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                   |
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| <p>(c) (i)</p> <p>After 1.5 seconds <math>x = 60</math> and <math>y = 2.25</math></p> $60 = 1.5V\cos\theta$ $V\cos\theta = 40 \text{ (1)}$ $2.25 = -5 \times 1.5^2 + 1.5V\sin\theta$ $13.5 = 1.5V\sin\theta$ $V\sin\theta = 9 \text{ (2)}$ <p>Dividing the two equations</p> $\frac{V\sin\theta}{V\cos\theta} = \frac{9}{40}$ $\tan\theta = \frac{9}{40}$ $\theta = \tan^{-1} \frac{9}{40} = 12^\circ 41'$ <p><math>\therefore</math> Golf ball has an angle of projection of <math>12^\circ 41'</math>.</p> | <p>3 marks: Correct answer.</p> <p>2 marks: Makes significant progress.</p> <p>1 mark: Sets up the two equations or shows some understanding.</p> |
| <p>(ii)</p> <p>Using equations (1) and (2)</p> $(V\sin\theta)^2 + (V\cos\theta)^2 = 9^2 + 40^2$ $V^2(\sin^2\theta + \cos^2\theta) = 9^2 + 40^2$ $V = \sqrt{9^2 + 40^2}$ $V = 41 \text{ ms}^{-1}$ <p><math>\therefore</math> Speed of the gold ball is <math>41 \text{ ms}^{-1}</math></p>                                                                                                                                                                                                                    | <p>1 Mark: Correct answer.</p>                                                                                                                    |
| <p>(d) (i)</p> $\frac{1}{100} \left( \frac{1}{P} + \frac{1}{100-P} \right) = \frac{1}{100} \left( \frac{100-P+P}{P(100-P)} \right) = \frac{1}{P(100-P)}$                                                                                                                                                                                                                                                                                                                                                     | <p>1 mark: Correctly shows the result</p>                                                                                                         |

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| <p>(ii)</p> $\frac{dP}{dt} = P(100 - P)$ $\frac{1}{P(100 - P)} dP = dt$ $\int \frac{1}{P(100 - P)} dP = \int dt$ $\int \frac{1}{100} \left( \frac{1}{P} + \frac{1}{100 - P} \right) dP = \int dt$ $\int \left( \frac{1}{P} + \frac{1}{100 - P} \right) dP = 100 \int dt$ $\log_e P - \log_e (100 - P) = 100t + c$ $t = 0, P = 10 \Rightarrow \log_e 10 - \log_e 90 = c$ $c = \log_e \frac{1}{9}$ $\log_e \left( \frac{P}{100 - P} \right) = 100t + \log_e \frac{1}{9}$ $\left( \frac{P}{100 - P} \right) = e^{100t + \log_e \frac{1}{9}}$ $\frac{P}{100 - P} = \frac{1}{9} e^{100t} \quad \Rightarrow \quad \frac{100 - P}{P} = 9e^{-100t}$ $9P = 100e^{100t} - Pe^{100t}$ $9P + Pe^{100t} = 100e^{100t}$ $P = \frac{100e^{100t}}{9 + e^{100t}}$ $\frac{100}{P} - 1 = 9e^{-100t}$ $\frac{100}{P} = 1 + 9e^{-100t}$ $P = \frac{100}{1 + 9e^{-100t}}$ | <p>3 marks: Correct answer</p> <p>2 marks: Integrates correctly without finding the constant <u>OR</u> Integrates correctly, finds c, but leaves as <math>t = f(P)</math>.</p> <p>1 mark: Makes some progress. I.e. separates the differential equation and attempts to integrate.</p>     |
| <p>(e)</p> <p>From the table, <math>f(x) = g^{-1}(x)</math> and so <math>f(-1) = g^{-1}(-1) = 0</math>.</p> $f'(-1) = \frac{1}{g'(f(-1))}$ $= \frac{1}{g'(0)}$ $= \frac{1}{\frac{1}{2}}$ $= 2$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>3 marks: Gives the correct solution</p> <p>2 marks:</p> <p>Determines <math>f(-1) = g^{-1}(-1) = 0</math> AND</p> $f'(-1) = \frac{1}{g'(f(-1))} \dots\dots\dots$ <p>1 mark:</p> <p>Determines <math>f(-1) = g^{-1}(-1) = 0</math> OR</p> $f'(-1) = \frac{1}{g'(f(-1))} \dots\dots\dots$ |